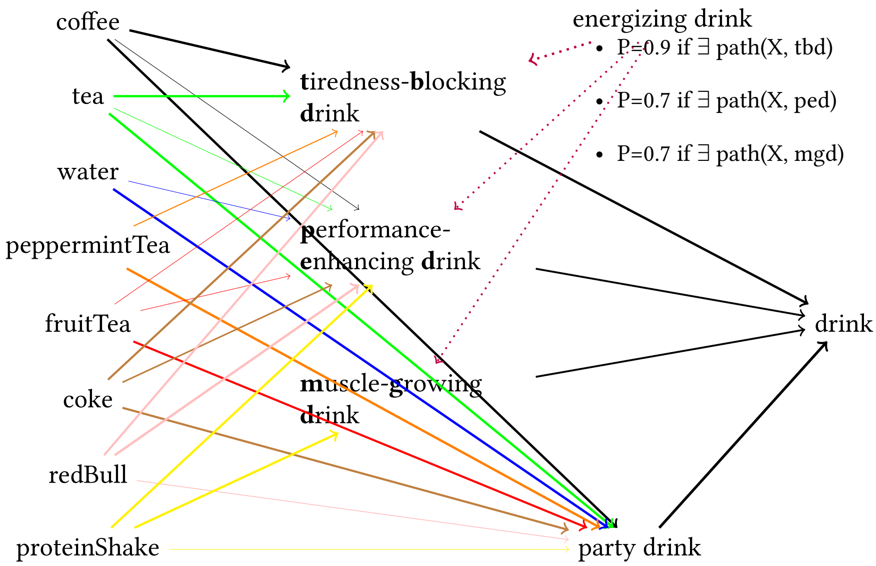


Figure 2: Linking word meaning to concepts in the encyclopaedia

5.3 Adjusting explicatures and implicatures in a Bayesian network

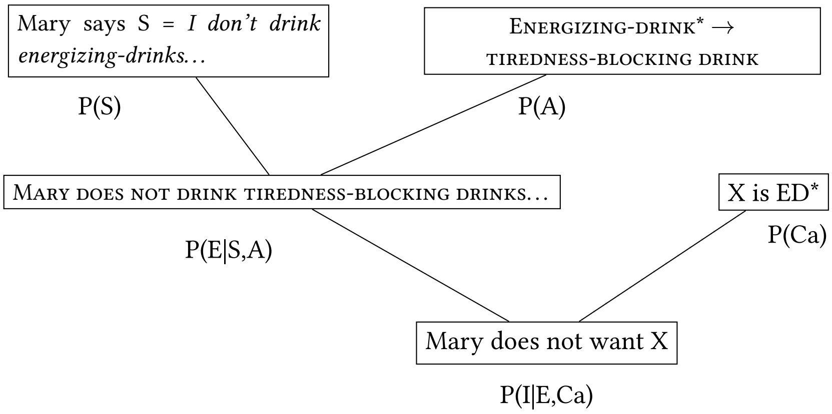
The parallel adjustment process of implicatures and explicatures can be modelled in a Bayesian network as indicated in figure 3. In our example, the implicature MARY DOES NOT WANT COFFEE depends on the one hand on the likelihood of the explicature MARY DOES NOT DRINK TIREDNESS-BLOCKING DRINKS AFTER 6PM being true, and on the likelihood of the implicit premise COFFEE IS A TIREDNESS-BLOCKING DRINK being true, on the other. Moreover, the explicature MARY DOES NOT DRINK TIREDNESS-BLOCKING DRINKS AFTER 6PM depends on the likelihood of Mary having said "I do not drink energizing drinks at this time of the day" and on the likelihood of the word "energizing drink" communicating the ad-hoc concept TIREDNESS-BLOCKING DRINK.[19]

5.4 Evaluating interpretive hypotheses

We have seen in section 3.4 that the communicative principle of relevance justifies a heuristic procedure that stops at the first easily accessible interpretive hypothesis that satisfies the audience's relevance expectations. The communicative principle of relevance entails that the interpretation found by this procedure is optimally relevant to the audience. In the previous section we have argued that in a probabilistic formalisation of relevance theory, we should capture the effects of relevance indirectly by considering the divergence of the probability distributions over sets of manifest propositions before

concepts created on the spot for the given occassion and not persistently stored in memory (Sperber and Wilson (1998), also published as chapter 2 of Wilson and Sperber (2012, pp. 31–46)). Since our main purpose is to model implicature comprehension, we do not address this issue. Notice that this limitation is also inherent in Erk and Herbelot (2023-11-30), who focus on modelling concepts that individuals persistently mentally represent.

[19]We are abstracting away from other factors entering into establishing the link between the sentence uttered and its conceptual representation.



S: description of the **S**entence uttered
A: pragmatic concept **A**djustment
E: **E**xplicature
Ca: **C**ontextual **a**ssumption; implicit premise
I: **I**mplicature; implicated conclusion

Figure 3: Parallel adjustment of implicatures and explicatures as a Bayesian network

and after processing utterances. The question arising at this point is what this means for modelling the process of finding the winning interpretive hypothesis.

In order to answer this question, we will procede as follows: we will construct various interpretive hypotheses *Int 1, Int 2, Int 3, Int 4, Int 5* differing in specific properties. We will then discuss what effects these properties have on the KL-divergence between the audience's prior estimates of probability/manifestness of propositions relating to the expected cognitive effects and the change in this probability distribution affected by the utterance under the respective interpretive hypothesis. We hope to show that when two interpretive hypotheses differing only in this KL measure, the hypothesis with the highest KL measure is the one that relevance theory predicts to be the optimally relevant one. However, not all interpretive hypotheses differ in their KL measure. We demonstrate that in such cases a comparison of the size of the Herbrand base of the probabilistic program – that is, the set of facts listed in the knowledge-base section of the program – allows to pick out the less costly to process hypothesis.[20]

(24) Interpretive hypotheses and their properties

a. *Int 1:* Optimally relevant interpretation: conveys a strong and various weaker implicatures satisfying relevance expectations about what beverage Mary is likely not to drink in addition to coffee. No superfluous context. Implementation provided in appendix B.

b. *Int 2:* Relevance expectations as in Int 1, but only enough contextual assumptions to satisfy only one implicit conclusion about Mary not drinking

[20] Apart from cases of genuine amgiuity in interpretation or accidental relevance. An example for the former may be Mozart's comment to his rival Salieri: *I didn't know that music like this is possible.* () An example for the latter case may be Margret Thatcher's comment *I always treat other people's money as if it were my own.* Relevance theory makes no prediction about how audiences settle on an interpretation in such cases.

coffee. Does not satisfy extant relevance expectations.

c. *Int 3:* Allows the same inferences to be drawn as in Int 1, satisfying the same relevance expectations as Int 1, but has additional superfluous encyclopaedic knowledge. This is supposed to be less relevant than Int 1 due to increased processing effort.

d. *Int 4:* Contains the same encyclopaedic knowledge about beverages at parties as Int 1, but adds another knowledge graph that allows inferences about whether Mary may be a particularly health-conscious person when not drinking tiredness-blocking drinks after 6PM. However, relevance expectations remain the same as in Int 1, so the extension of the space of possible available inferences does not improve relevance expectation satisfaction.

e. *Int 5:* Same as Int 3: but now the relevance expectations include questions about Mary's health-consciousness (and what this may mean not only for drinks, but perhaps also for foods that Peter plans to offer later at the party).

5.4.1 Interpretive hypothesis Int 1

This interpretive hypothesis assumes that it is highly relevant to Peter to know whether Mary wants to accept his offer of a cup of coffee, and that it would be relevant to him to know that in the case Mary declines coffee, what alternative he may be able to offer with less chance of refusal. We model these relevance expectations as a set of queries of the form `wantsNotDrink(mary,X)` which can be seen as a placeholder for propostions of the form MARY DOES NOT WANT TO DRINK X, where X is a variable standing for the various beverages that Peter happens to offer. We assume that this list includes coffee, coke, fruit tea, peppermint tea, tea, water, and also some untypical beverages that Peter may be able to offer: protein shakes and Red Bull (energy drinks). We further assume that Peter has no particular knowledge of Mary's preferences, so prior to Mary's utterance, none of the assumptions that Mary does not want to drink any of the available beverages are highly manifest to him, the likelihood that Peter mentally represents any of them as a true proposition is about chance level, perhaps assuming that people are slightly more likely to drink coffee than peppermint tea of fruit tea at 6PM, and less likely to want to drink a non-typical party drink. This is captured in the probability distribution in the column P (prior).

Mary's utterance provides Peter with the evidence that she does not energizing drinks after 6PM. Passing this evidence to the ProbLog program listed in B means setting the probability that Mary does not want to drink any X that is connected to the category ENERGIZING DRINK* to true (P=1.0). Given the encyclopaedic knowledge graphs described in 5.1 and 5.2, this will change the likelihood that Peter is able to mentally represent the respective propositions in the ways indicated in the column P (after). The divergence between the probability distribution P (prior) and P (after) is calculated in the column KLD. The KL measure in this column is calculated with the `kl_div` function of the Python library `scipy.special`. The function is defined as follows:

$$kl\_div(x,y) = \begin{cases} x * log(x/y) - x + y & x > y, y > 0 \\ y & x = 0, y \geq 0 \\ \infty & otherwise \end{cases}$$